

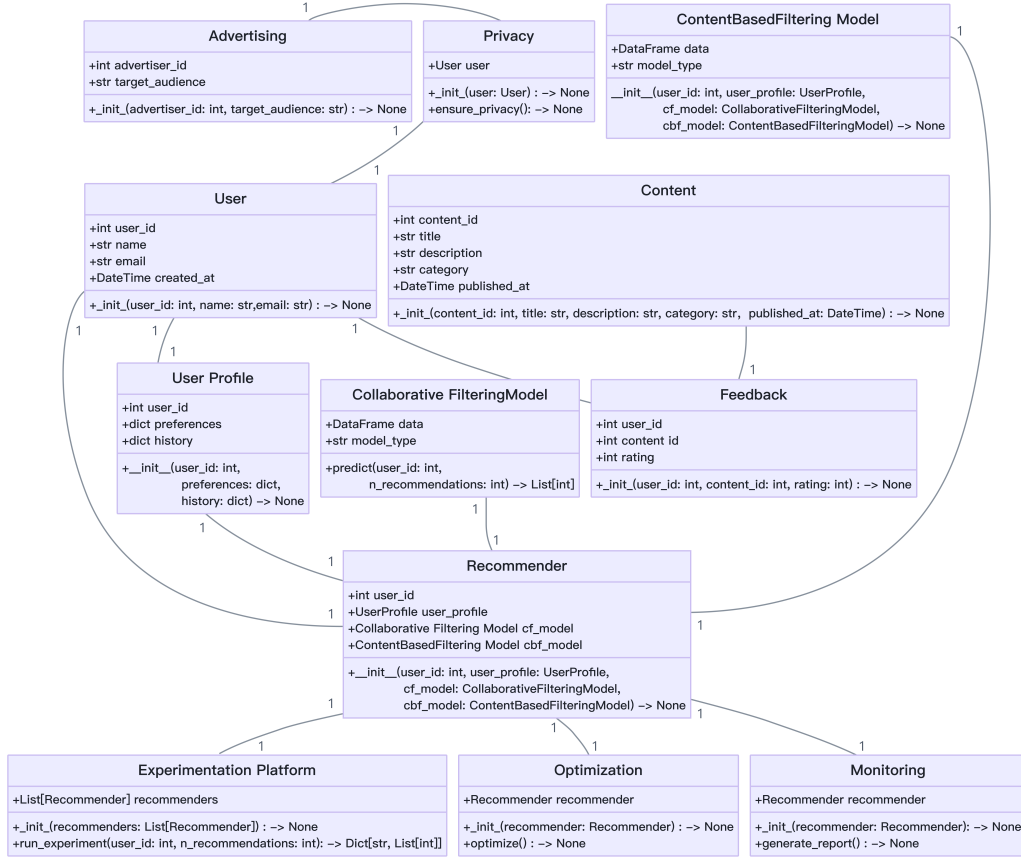


Figure 11: The system interface design for “recommendation engine development” is generated by the *architect* agent (zoom in for a better view).

E MORE DISCUSSIONS

E.1 DEEP-SEATED CHALLENGES

MetaGPT also alleviates or solves these challenges with its unique designs:

Use Context Efficiently Two sub-challenges are present. First, unfolding short natural language descriptions accurately to eliminate ambiguity. Second, maintaining information validity in lengthy contexts, enables LLMs to concentrate on relevant data without distraction.

Reduce Hallucinations Using LLMs to generate entire software programs faces code hallucination problems—including incomplete implementation of functions, missing dependencies, and potential undiscovered bugs, which may be more serious. LLMs often struggle with software generation due to vague task definitions. Focusing on granular tasks like requirement analysis and package selection offers guided thinking, which LLMs lack in broad task solving.

E.2 INFORMATION OVERLOAD

In MetaGPT, we use a global message pool and a subscription mechanism to address “information overload,” which refers to the problem of receiving excessive or irrelevant information. This issue is dependent on specific applications. MetaGPT employs a message pool to streamline communication, ensuring efficiency. Additionally, a subscription mechanism filters out irrelevant contexts, enhancing the relevance and utility of the information. This design is particularly crucial in soft-

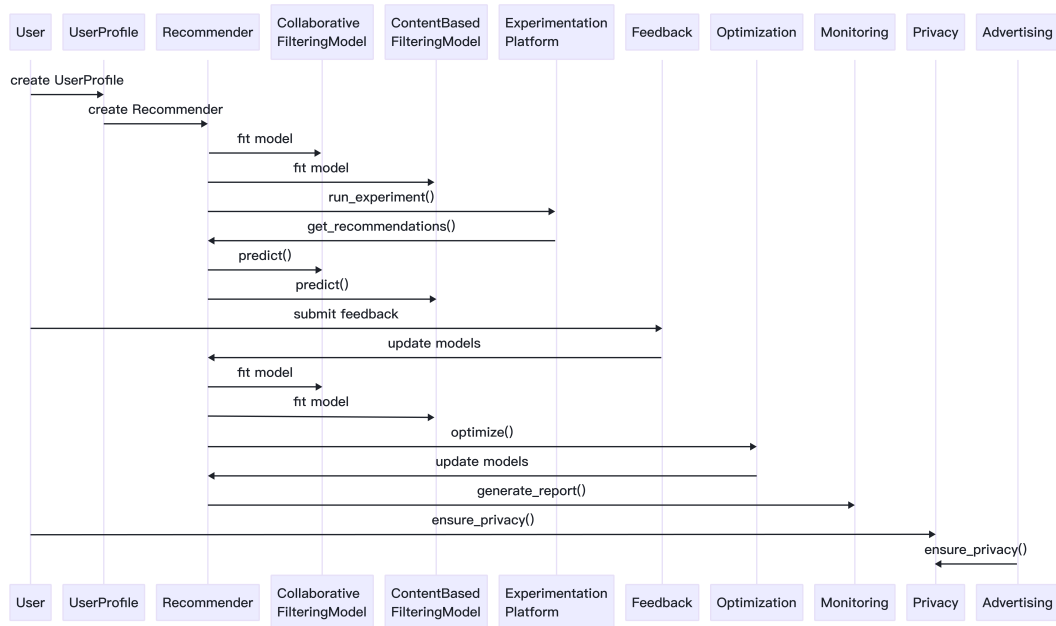


Figure 12: The program call flow for “recommendation engine development” generated by the *architect* agent (**zoom in for a better view**).

ware design scenarios and standard operating procedures (SOPs) where effective communication is essential.

Table 8: Examples of SoftwareDev dataset.

Task ID	Task	Prompt
0	Snake game	Create a snake game.
1	Brick breaker game	Create a brick breaker game.
2	2048 game	Create a 2048 game for the web.
3	Flappy bird game	Write p5.js code for Flappy Bird where you control a yellow bird continuously flying between a series of green pipes. The bird flaps every time you left click the mouse. If it falls to the ground or hits a pipe, you lose. This game goes on indefinitely until you lose; you get points the further you go.
4	Tank battle game	Create a tank battle game.
5	Excel data process	Write an excel data processing program based on streamlit and pandas. The screen first shows an excel file upload button. After the excel file is uploaded, use pandas to display its data content. The program is required to be concise, easy to maintain, and not over-designed. It uses streamlit to process web screen displays, and pandas is sufficient to process excel reading and display. Please make sure others can execute directly without introducing additional packages.
6	CRUD manage	Write a management program based on the crud addition, deletion, modification and query processing of the customer business entity. The customer needs to save this information: name, birthday, age, sex, and phone. The data is stored in client.db, and there is a judgement whether the customer table exists. If it doesn't, it needs to be created first. Querying is done by name; same for deleting. The program is required to be concise, easy to maintain, and not over-designed. The screen is realized through streamlit and sqlite—no need to introduce other additional packages.
7	Music transcriber	Develop a program to transcribe sheet music into a digital format; providing error-free transcribed symbolized sheet music intelligence from audio through signal processing involving pitch and time slicing then training a neural net to run Onset Detected CWT transforming scalograms to chromagrams decoded with Recursive Neural Network focused networks.
8	Custom press releases	Create custom press releases; develop a Python script that extracts relevant information about company news from external sources, such as social media; extract update interval database for recent changes. The program should create press releases with customizable options and export writings to PDFs, NYTimes API JSONs, media format styled with interlink internal fixed character-length metadata.
9	Gomoku game	Implement a Gomoku game using Python, incorporating an AI opponent with varying difficulty levels.
10	Weather dashboard	Create a Python program to develop an interactive weather dashboard.

Table 9: **Additional results of pure MetaGPT w/o feedback on SoftwareDev.** Averages (Avg.) of 70 tasks are calculated and 10 randomly selected tasks are included. '#' denotes 'The number of', while 'ID' is 'Task ID'.

ID	Code statistics			Doc statistics			Cost statistics			Cost of revision	Code executability	
	#code files	#lines of code	#lines per code file	#doc files	#lines of doc	#lines per doc file	#prompt tokens	#completion tokens	time costs			money costs
0	5.00	196.00	39.20	3.00	210.00	70.00	24087.00	6157.00	582.04	\$ 1.09	1. TypeError	4
1	6.00	191.00	31.83	3.00	230.00	76.67	32517.00	6238.00	566.30	\$ 1.35	1. TypeError	4
2	3.00	198.00	66.00	3.00	235.00	78.33	21934.00	6316.00	553.11	\$ 1.04	1. lack @app.route('/')	3
3	5.00	164	32.80	3.00	202.00	67.33	22951.00	5312.00	481.34	\$ 1.01	1. PNG file missing 2. Compile bug fixes	2
4	6.00	203.00	33.83	3.00	210.00	70.00	30087.00	6567.00	599.58	\$ 1.30	1. PNG file missing 2. Com-pile bug fixes 3. pygame.surface not initialize	3
5	6.00	219.00	36.50	3.00	294.00	96.00	35590.00	7336.00	585.10	\$ 1.51	1. dependency er-ror 2. ModuleNotFoundError	4
6	4.00	73.00	18.25	3.00	261.00	87.00	25673.00	5832.00	398.83	\$ 0.90	0	4
7	4.00	316.00	79.00	3.00	332.00	110.67	29139.00	7104.00	435.83	\$ 0.92	0	4
8	5.00	215.00	43.00	3.00	301.00	100.33	29372.00	6499.00	621.73	\$ 1.27	1. tensorflow ver-sion error 2. model training method not implement	2
9	5.00	215.00	43.00	3.00	270.00	90.00	24799.00	5734.00	550.88	\$ 1.27	1. dependency er-ror 2. URL 403 er-ror	3
10	3.00	93.00	31.00	3.00	254.00	84.67	24109.00	5363.00	438.50	\$ 0.92	1. dependency er-ror 2. missing main func.	4
Avg.	4.71	191.57	42.98	3.00	240.00	80.00	26626.86	6218.00	516.71	\$ 1.12	0.51 (only consider item scored 2, 3 or 4)	3.36